

End-to-End Process · Plan-to-Produce (P2Pr)

HITL Design

PCF 4.1 Plan for and Align Supply Chain Resources · PCF 4.3 Produce / Assemble / Test Product · PCF 4.4 Manage Logistics and Warehousing · Functions: Engineering · Materials · Manufacturing · Supplier

P2Pr HITL Design Summary

P2Pr Aggregate H% · M3	Decision Gates	Knowledge Steps	Exception Flows	Δ M1 → M3
53% Human	3 · 80% H	5 · 65% H	4 · 82% H	-25%

H% by Pattern Type at Each Maturity Level

Pattern	Nature of Work	M1 Undesigned	M2 Emerging	M3 ★ Designed	M4 Optimized	M5 Leading
Decision	Judgment & Authority	92%	88%	80%	72%	65%
Knowledge	Synthesis & Interpretation	85%	78%	65%	52%	42%
Document	Content Generation	72%	58%	40%	28%	18%
Transaction	Rules-Based Processing	60%	48%	35%	22%	12%
Exception	Non-Standard Resolution	95%	92%	82%	75%	68%

HITL Impact Analysis

H% = Human involvement at each maturity level · M3 = HITL Design Intent

Process Step	PCF	Pattern	M1 H%	M3 H% ★	Δ M1→M3	AI Opportunity
P2Pr End-to-End Process · L1 Aggregate	4.1+4.3	L1 AGG	78%	53%	-25%	Significant
Engineering · PCF 4.3 — BOMs, Routings, and Product Specifications						
↳ Develop Bill of Materials	4.3.1	Document	72%	40%	-32%	High
↳ Define Production Routings	4.3.1	Knowledge	85%	65%	-20%	High
↳ Manage Engineering Change Orders	4.3.1	Decision	92%	80%	-12%	Retain Human
↳ → Manufacturing (Release)	4.3.1	Transaction	60%	35%	-25%	High
Materials · PCF 4.1 + 4.4 — Supply Chain Planning, Inventory, and Logistics						
↳ Develop Supply Chain Strategy	4.1.1	Knowledge	85%	65%	-20%	High
↳ Develop Demand Forecast	4.1.2	Knowledge	85%	65%	-20%	High
↳ Develop Inventory Plan	4.1.3	Decision	92%	80%	-12%	Retain Human
↳ Develop Production Plan	4.1.4	Decision	92%	80%	-12%	Retain Human
↳ Manage Supply Chain Risk	4.1.5	Knowledge	85%	65%	-20%	High
↳ Receive and Store Materials	4.4.1	Transaction	60%	35%	-25%	High
↳ Manage Inventory	4.4.2	Transaction	60%	35%	-25%	High
↳ → Purchasing (P2P)	4.1.4	Transaction	60%	35%	-25%	High
↳ Planning Exception	4.1.2	Exception	95%	82%	-13%	Retain Human

Manufacturing · PCF 4.3 — Production Scheduling, Execution, and Quality						
↳ Develop Production Schedule	4.3.2	Knowledge	85%	65%	-20%	High
↳ Manage Work Orders	4.3.3	Transaction	60%	35%	-25%	High
↳ Execute Production	4.3.4	Transaction	60%	35%	-25%	High
↳ Perform Quality Inspection	4.3.5	Decision	92%	80%	-12%	Retain Human
↳ Manage Non-Conformance	4.3.6	Exception	95%	82%	-13%	Retain Human
↳ Record Production Output	4.3.7	Transaction	60%	35%	-25%	High
↳ → Materials (Inventory Update)	4.3.7	Transaction	60%	35%	-25%	High
↳ Production Exception	4.3.4	Exception	95%	82%	-13%	Retain Human
External Participant · Supplier / Contract Manufacturer						
↳ Receive Purchase Order	Ext	Transaction	60%	35%	-25%	High
↳ Confirm Materials Availability	Ext	Transaction	60%	35%	-25%	High
↳ Deliver Materials / Components	Ext	Transaction	60%	35%	-25%	High
↳ → Materials (Goods Receipt)	Ext	Transaction	60%	35%	-25%	High
↳ Supplier Exception	Ext	Exception	95%	82%	-13%	Retain Human

HITL Design Insight — P2Pr has the largest AI convergence gap of any module at -25% — driven by high transaction volume across production execution, work order management, materials receiving, and inventory management. Engineering anchors the process with BOMs and routings — the irreducible human judgment that defines what gets made and how. Materials is the operational hub: planning, inventory, logistics, and the handoff to Purchasing (P2P). Manufacturing executes, inspects, and closes the loop back to Materials. P2Pr aggregate at M3: 53% Human / 47% AI.